

Table 1: Cost-efficiency of open-source vs. frontier LLMs for smart-contract vulnerability detection. Ensemble type-aware recall with 95% Wilson confidence intervals and measured API cost, on a DeFi business-logic corpus ($n=27$ vulnerabilities) and a SmartBugs-curated classical-pattern subset ($n=36$ contracts). Claude Fable 5 refuses the task across three framings (safety guardrail) and is reported as REFUSED. All model confidence intervals overlap within each corpus, i.e. the models are statistically comparable, while cost differs by 3–30 $\times$.

Model	Type	DeFi ($n=27$)			Classical ($n=36$)		
		Recall	95% CI	Cost	Recall	95% CI	Cost
DeepSeek-reasoner	OSS hosted	92.6%	[77, 98]	\$0.20	86.1%	[71, 94]	\$0.32
GPT-5	frontier	88.9%	[72, 96]	\$0.54	94.4%	[82, 98]	\$0.94
Claude Sonnet 4.6	frontier	85.2%	[68, 94]	\$0.19	97.2%	[86, 100]	\$0.61
GPT-4o	frontier	37.0%	[22, 56]	\$0.05	97.2%	[86, 100]	\$0.11
qwen3-coder:30b	local (\$0)	63.0%	[44, 78]	\$0.00	94.4%	[82, 98]	\$0.00
Claude Fable 5	premium	REFUSED			REFUSED		

Table 2: Open-weight vs. frontier LLMs on **EVMBench** (40 real DeFi-audit codebases, 120 HIGH-severity vulnerabilities; ground truth = professional audit findings), each a *single* stored run through the same MIESC harness (`evmbench_eval.py` v5.3.1), so the rows are run-count matched. Recall with 95% Wilson intervals, precision, F1, and per-audit cost. At matched run count the open-weight DeepSeek trails the frontier on recall but leads on precision and F1 at 15–69 $\times$ lower cost; a 3-run DeepSeek self-ensemble ($\approx$ \$0.24/audit) reaches 81.7%, within the frontier single-run range.

Model	Type	Recall	95% CI	Prec.	F1	\$/audit
Claude Sonnet 4.6	frontier	82.5%	[75, 88]	34.0%	48.2%	5.50
GPT-5	frontier	77.5%	[69, 84]	42.7%	55.0%	3.00
GPT-4o	frontier	73.7%	[65, 81]	20.0%	31.5%	1.20
DeepSeek-reasoner	OSS (MIT)	70.8%	[62, 78]	88.5%	78.7%	0.08*
qwen2.5-coder:32b	local	59.2%	[50, 68]	30.1%	39.9%	0.00
static-only	baseline	18.3%	[12, 26]	14.5%	16.2%	0.00
4-frontier ensemble (union)		92.5%	—	—	—	7.00

*DeepSeek cost estimated from \$0.55/\$2.19 per-MTok pricing; the harness does not capture per-call adapter tokens.